

JEE MAIN REVISION MASTER: SEQUENCES AND SERIES 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Arithmetic & Geometric Progressions (AP, GP, AGP)	11 PYQs	Progressions
Standard	AM-GM-HM Inequality & Telescoping Series	2 PYQs	Inequalities & Sigma

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Arithmetic & Geometric Progressions (AP, GP, AGP)	$S_n = \frac{n}{2}[2a + (n-1)d]$, $S_n = \frac{a(1-r^n)}{1-r}$, $S_{\infty} = \frac{a}{1-r}$; $(r < 1)$, $T_n = S_n - S_{n-1}$
AM-GM-HM Inequality & Telescoping Series	$\frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b}$, $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum n^3 = [\sum n]^2$, $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Arithmetic & Geometric Progressions (AP, GP, AGP) (High Yield — 11 PYQs)

Core Principle: nth term and sum formulas for AP and GP, infinite geometric series convergence, and Arithmetico-Geometric series summation.

■■ **Caution / JEE Pitfall:** The formula $S_{\infty} = \frac{a}{1-r}$ is valid ONLY when $|r| < 1$. If $|r| \geq 1$, the infinite series diverges.

■ **Speed Trick:** To choose 3 terms in AP: $a - d, a, a + d$ (sum eliminates d). In GP: $a/r, a, ar$ (product eliminates r).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If we insert two numbers between $\sqrt{2}$ and 4 so that the resulting sequence is in G.P, then the inserted numbers in the order are..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "If 4th, 10th and 16th terms of a G.P. are x, y and z respectively, then..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "If S_n stands for sum to n -terms of a GP with a as the first term and r as the common ratio, then $S_n : S_{2n}$ is..." [Ans: N/A]

3.2 AM-GM-HM Inequality & Telescoping Series (Standard Yield — 2 PYQs)

Core Principle: Arithmetic, geometric, and harmonic mean inequalities for positive numbers, sum of first n natural powers, and partial fraction telescoping.

■ **Caution / JEE Pitfall:** $AM \geq GM$ equality holds IF AND ONLY IF all participating positive numbers are equal ($a_1 = a_2 = \dots = a_n$).

■ **Speed Trick:** Whenever asked for the minimum value of a sum of positive terms whose product is constant (e.g. $x + (1/x)$), use $AM \geq GM$.

Representative Source PYQs:

- [2023 Shift] Q#1 (MCQ): " n th term of the series $1 + (3/7) + \frac{57^2}{17^2} + \dots$ is..." [Ans: N/A]
- [2021 Shift] Q#1 (MCQ): "If the middle term of the AP is 300, then the sum of its first 51 terms is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If we insert two numbers between $\sqrt[3]{2}$ and 4 so that the resulting sequence is in G.P, then the inserted numbers in the order are

- (A) $8, \sqrt[3]{2}$
- (B) $2, \sqrt[3]{8}$
- (C) $\sqrt[3]{8}, 2$
- (D) $\sqrt[3]{2}, 8$

Correct Answer: None

Solution Analysis: The sequence after inserting two numbers will be $\sqrt[3]{2}, x, y, 4$. Since these four terms are in G.P., let the common ratio be r . The first term is $\sqrt[3]{2}$ and the fourth term is 4. For a G.P., the fourth term is ar^3 . So, $\sqrt[3]{2}r^3 = 4$
 $r^3 = \frac{4}{\sqrt[3]{2}}$ Now, $\frac{4}{\sqrt[3]{2}} = \frac{2^2 \cdot 2^{1/3}}{2^{1/3}} = 2^{5/3}$ So, $r^3 = 2^{5/3}$ Taking cube root, $r = 2^{5/9}$ Now the terms are:
 $\sqrt[3]{2}, \sqrt[3]{2} \times \sqrt[3]{2}, 2 \times \sqrt[3]{2}, 4$ So, $x=2$ and $y=\sqrt[3]{8}$ Hence, the inserted numbers in order are $2, \sqrt[3]{8}$ Correct option: Option B.

Benchmark Problem 2: KCET 2023 (Q#1)

n th term of the series $1 + (3/7) + \frac{57^2}{17^2} + \dots$ is

- (A) $\frac{2n+17^n}{17^n}$
- (B) $\frac{2n-17^n}{17^n}$
- (C) $\frac{2n+17^{n-1}}{17^n}$
- (D) $\frac{2n-17^{n-1}}{17^n}$

Correct Answer: None

Solution Analysis: We have, $1 + (3/7) + (5/7^2) + (7/7^3) + \dots$ (1/70) + (3/7) + (5/7^2) + (7/7^3) + ... Numerator are in AP So, T_n of AP = $1 + (n-1) \cdot 2 = 2n-1$ Denominator are in GP So, T_n of GP = 7^{n-1} T_n of given series = $\frac{2n-17^{n-1}}{17^n}$